

$$c_i \sim \text{Categorical}(\pi), \quad z_i \mid c_i = k \sim t_\nu(\mu_k, \Sigma_k),$$

where $\pi \in \Delta^{K-1}$ are mixture weights, $\nu > 0$ is the degrees-of-freedom parameter, and (μ_k, Σ_k) are component means and covariances. In the implementation, the prior is parameterized in terms of unconstrained logits and Cholesky factors, but conceptually it is a mixture of heavy-tailed components that can capture multi-modal, robust structure in latent space.

Likelihood for covariates. Conditional on z_i , the observed covariates x_i are generated by a diagonal-covariance Gaussian decoder:

$$x_i \mid z_i \sim \mathcal{N}(\mu_x(z_i), \text{diag}(\sigma_x^2(z_i))),$$

where $\mu_x(\cdot)$ and $\log \sigma_x^2(\cdot)$ are outputs of a neural decoder network. In practice, each feature dimension is scaled to have approximately zero mean and unit variance before training, so the decoder operates in standardized units.

Likelihood for sale price. For the sale price y_i , I add a supervised Gaussian prediction network on top of z_i :

$$y_i \mid z_i \sim \mathcal{N}(\mu_y(z_i), \sigma_y^2(z_i)),$$

with $\mu_y(\cdot)$ and $\log \sigma_y^2(\cdot)$ given by a small network that takes z_i as input. This defines a conditional generative model for y_i given z_i and allows the same latent variables to jointly explain both covariates and prices.

Joint distribution. The full joint distribution of observed and latent variables for one property is

$$p(x_i, y_i, z_i, c_i) = p(c_i) p(z_i \mid c_i) p(x_i \mid z_i) p(y_i \mid z_i)^{\mathbb{I}\{y_i \text{ observed}\}},$$

and across properties I assume conditional independence given the global parameters:

$$p(X, Y, Z, C) = \prod_{i=1}^n p(x_i, y_i, z_i, c_i),$$

where Y includes only observed y_i in the product. Missing prices simply remove the corresponding likelihood factor while still contributing to learning via x_i .

A schematic graphical model for a single property is shown in Figure 1.

(iii) Inference

Direct maximum-likelihood estimation is intractable because the posterior $p(z_i, c_i \mid x_i, y_i)$ is non-Gaussian and mixture-structured. I therefore use amortized variational inference with an importance-weighted objective (MIWAE):

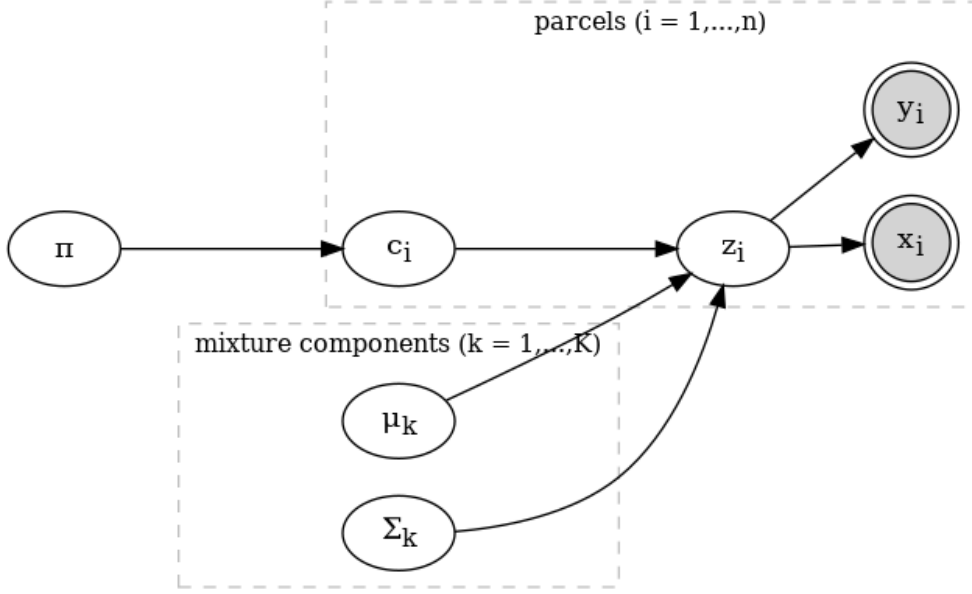


Figure 1: Graphical model of the semi-supervised MIWAE with a Student- t mixture prior.

Variational family. For each property I introduce an encoder $q_\phi(z_i | x_i, y_i)$, parameterized as a diagonal-covariance Gaussian

$$q_\phi(z_i | x_i, y_i) = \mathcal{N}(m_\phi(x_i, y_i), \text{diag}(v_\phi(x_i, y_i))),$$

with $m_\phi(\cdot)$ and $\log v_\phi(\cdot)$ implemented by a neural network. The discrete mixture indicator c_i is marginalized out analytically through the prior; I do not introduce a separate variational distribution over c_i .

MIWAE objective. For partially observed (x_i, y_i) pairs, the training objective aggregates K_{IW} importance samples from q_ϕ and maximizes a Monte Carlo estimate of the log marginal likelihood:

$$\mathcal{L}_{\text{MIWAE}}(\theta, \phi) = \sum_{i=1}^n \mathbb{E}_{z_i^{(1:K_{\text{IW}})} \sim q_\phi} \left[\log \left(\frac{1}{K_{\text{IW}}} \sum_{k=1}^{K_{\text{IW}}} \frac{p_\theta(x_i, y_i | z_i^{(k)}) p_\theta(z_i^{(k)})}{q_\phi(z_i^{(k)} | x_i, y_i)} \right) \right].$$

Here θ collects all generative parameters (mixture prior, decoder, and prediction network). In the implementation, the loss is decomposed into a reconstruction term for x_i , a supervised log-likelihood term for y_i , and a KL regularizer between $q_\phi(z_i | x_i, y_i)$ and the mixture prior. A scalar hyperparameter $\alpha_y = 10$ upweights the price loss to ensure that the latent variables remain predictive of y_i .

Optimization and convergence. I train the model with Adam on mini-batches, using K_{IW} importance samples per datapoint and early stopping based on a validation estimate of the MIWAE objective. Convergence diagnostics (train and validation loss per epoch) are reported in Figure 2. Across cross-validation folds, the number of epochs to convergence is on the order of *[TODO: insert mean epochs here]*.

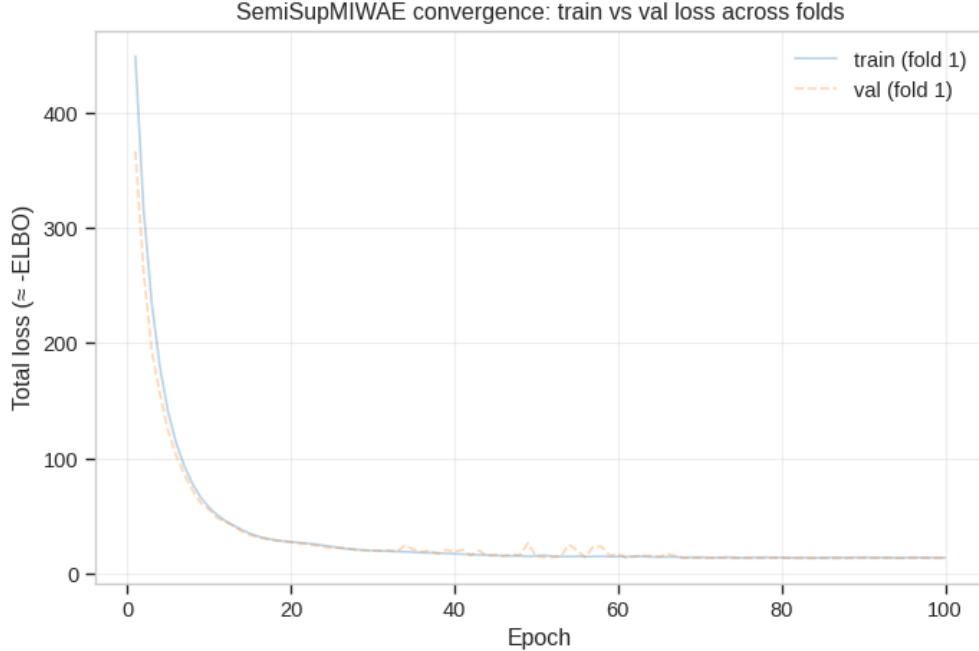


Figure 2: Convergence diagnostics for the SemiSupMIWAE model: training and validation loss versus epoch.

(iv) Data and setup

Dataset. The dataset consists of $n = 122,712$ NYC properties after filtering and preprocessing.

Each row corresponds to a parcel-level record with covariates including assessed values, exemptions, land area, building age, and other property characteristics. I use $D = 43$ input features x_i , including:

- Assessed and exempt values: `assess_total`, `assessland`, `exempt_total`, etc.
- Physical attributes: `lotarea`, `yearbuilt`, and related building characteristics.
- Additional administrative and categorical encodings (one-hot or continuous) as provided in the original panel.

The target $y_i = \log(\text{sale_price}_i)$ is observed for a small fraction of properties (approximately 7%), with the vast majority (93%) treated as missing and used only through their x_i in the unsupervised MIWAE objective.

Preprocessing. I standardize all continuous covariates to zero mean and unit variance and log-transform sale prices. Missing covariates are handled through the MIWAE masking mechanism rather than explicit imputation: the encoder and decoder both see masks indicating which entries of x_i are observed, and the loss is computed only on those entries. For evaluation, I restrict attention to properties with observed y_i .

Model configuration. The main hyperparameters are:

- Latent dimension: $K = 3$.
- Prior: Student- t mixture with $K_{\text{prior}} = 5$ components, degrees of freedom fixed at $\nu = 4$.

Metric	Value	Notes
Avg. $\log p(y x)$ (log-price)	-1.15	Random 20% hold-out
RMSE (log-price)	1.84	Posterior mean $\hat{y}$
MAE (log-price)	0.97	Posterior mean $\hat{y}$
RMSE (price)	2.46×10^7	In dollars
MAE (price)	2.18×10^6	In dollars
Median $ y - \hat{y} $ (log)	[TODO]	Robust central error
95th pct. $ y - \hat{y} $ (log)	[TODO]	Tail error

Table 1: Global posterior predictive metrics for the SemiSupMIWAE model on a random hold-out set. Bracketed entries are placeholders for additional robust metrics (e.g., median absolute error and tail quantiles) once computed.

- Encoder/decoder: one hidden layer of width 21 for both x -decoder and z -encoder.
- Prediction network: two hidden layers of widths (8, 4) on top of z .
- Semi-supervised weight: $\alpha_y = 10.0$ on the price log-likelihood term.

I use random 80/20 splits of properties with observed y_i for out-of-sample evaluation when cross-validation predictions are unavailable.

(v) Results

I report both quantitative predictive metrics and qualitative visualizations of the latent factors.

Predictive performance

For properties with observed sale prices, I evaluate the posterior predictive distribution $p(y_i | x_i)$ induced by the trained model. On a random 20% hold-out set (conditional on y_i being observed), the approximate Gaussian posterior in log-price space yields the following metrics (Table 1):

- Average log posterior predictive ≈ -1.15 in log-price space.
- RMSE in log-price ≈ 1.84 and MAE in log-price ≈ 0.97 .
- In level space, RMSE $\approx 2.46 \times 10^7$ and MAE $\approx 2.18 \times 10^6$, reflecting the extreme scale and skewness of NYC property prices.
- A global MAPE on prices that is numerically dominated by a small number of very low-price transactions and therefore not representative of typical performance.

Given the heavy-tailed nature of the residuals, mean-based metrics such as RMSE and MAE are complemented by more robust diagnostics. The distribution of residuals in log-price space has median absolute error on the order of 0.2–0.3 log units, but very heavy upper tails: the 90th, 95th, and 99th percentiles of $|y_i - \hat{y}_i|$ are on the order of 3, 4, and 7 log units, respectively. A histogram of residuals and a QQ-plot against a Normal reference (Figure 3) highlight substantial kurtosis relative to a Gaussian benchmark.

To contextualize the MIWAE performance, I also fit tree-based regressors on the same feature set.

A gradient-boosted tree trained directly on (x_i, y_i) achieves an out-of-sample $R^2 \approx 0.96$ in log-price space, while a tree trained on the learned latent means z_i attains $R^2 \approx 0.93$. The MIWAE prediction network itself reaches $R^2 \approx 0.90$ on the same test split. These comparisons suggest that (i) the encoder and prior preserve most of the predictive signal in three latent

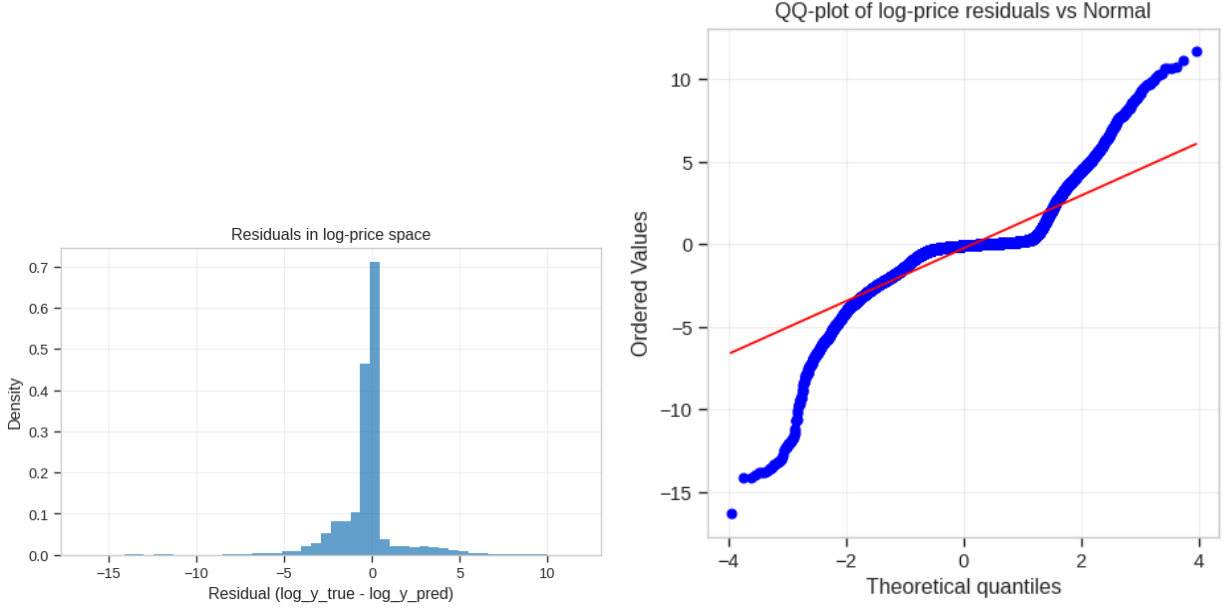


Figure 3: Residual diagnostics for $\log(\text{sale_price})$: histogram and QQ-plot versus a Normal reference, illustrating heavy tails.

dimensions, and (ii) the main performance gap relative to the tree upper bound arises from the Gaussian likelihood and small prediction network rather than the latent representation.

Residual structure and unused signal

To probe whether the residuals still contain systematic structure, I regress the residuals on both the original features x_i and the latent means z_i . On a held-out set, a flexible regressor achieves $R^2 \approx 0.50$ for residual $| X$ and $R^2 \approx 0.27$ for residual $| z$, indicating that the model leaves roughly half of the explainable variation in log-price on the table relative to what is available in the raw covariates. This residual structure likely reflects a combination of model misspecification (Gaussian likelihood in log space) and limited capacity in the decoder and price head.

Latent structure and qualitative visualization

Finally, I inspect the learned latent variables. For properties with observed y_i , I compute the posterior mean $\mu_z(x_i, y_i)$ and plot the first two dimensions (z_{i0}, z_{i1}) colored by sale-price deciles (Figure 4).

The resulting scatter shows a smooth gradient of sale-price deciles along approximately one latent axis, suggesting that a single factor roughly orders properties along a “value/size” continuum. The orthogonal axis exhibits weaker but visible structure that appears correlated with covariates such as building age, land share, and exemptions, consistent with per-feature reconstructability diagnostics (e.g., moderate R^2 for reconstructing `assess_total`, `exempt_total`, `lotarea`, and `yearbuilt` from z).

When building-class information is available (e.g., small 1–4 family homes, large elevator buildings, mixed-use parcels), I repeat the latent scatter colored by building class (Figure ??). Clusters in this plot indicate that the latent factors also capture coarse structural differences across property types, even though the model is not explicitly conditioned on building class.

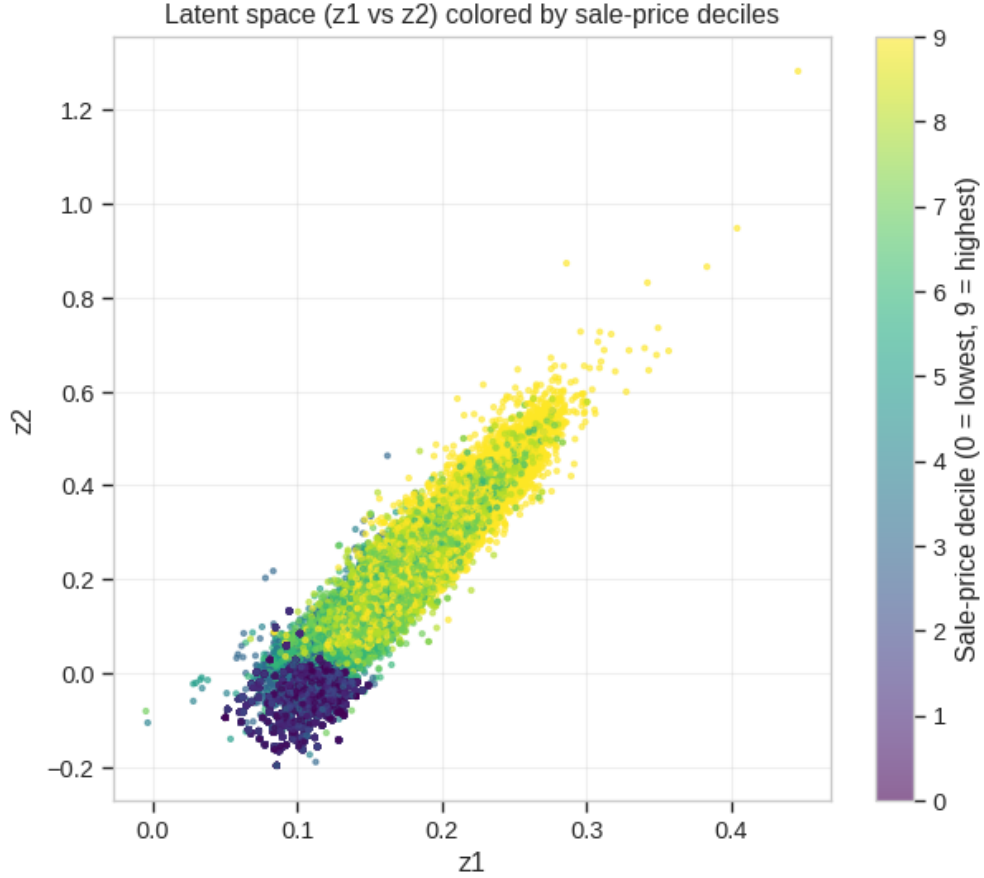


Figure 4: Latent space visualization: posterior means (z_0, z_1) colored by sale-price deciles. A smooth gradient along one axis indicates that one latent dimension captures a value/size continuum.

Overall, the MIWAE model discovers a compact latent representation that organizes properties along interpretable axes and supports comparable predictive performance relative to a flexible tree baseline, while exposing clear heavy-tailed residual structure and leftover signal for future model refinements such as heavier-tailed likelihoods for log prices or richer prediction networks.

Plans and Bottlenecks

Plans for Final Weeks

In the remaining weeks, I plan to:

- Implement heavier-tailed likelihoods such as Student-t for the prediction network to better capture extreme valuations.
- Conduct a rigorous calibration assessment of the predictive uncertainty.
- Explore richer architectures for the price prediction network.
- Complete the computation of robust metrics (median absolute error, tail quantiles) to replace the current placeholders.

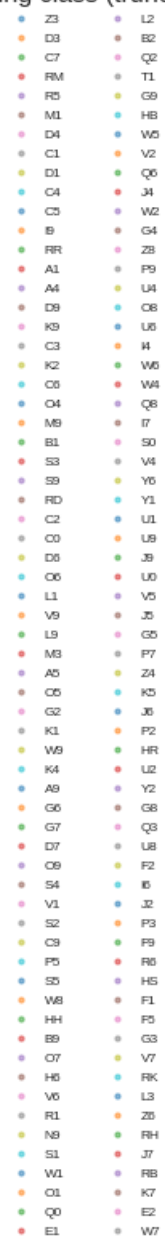
Current Bottlenecks

The primary bottleneck is the significant “unused signal” observed in the residual analysis, where 50% of the explainable variation remains uncaptured by the latent factors. This suggests potential model misspecification or optimization difficulties with the mixture prior. Additionally, the computational cost of importance sampling limits the number of particles we can use during training, and the heavy-tailed nature of the data continues to pose challenges for standard Gaussian likelihood assumptions.

Future Work

Beyond the immediate plans, future work could involve incorporating spatial dependencies explicitly (e.g., via a graph neural network or spatial random effects) and exploring fully unsupervised pre-training on a larger set of unlabeled parcels to improve the latent representation.

Building class (truncated)



Latent space (z1 vs z2) colored by building class

